Hwang 2022 Pancreatic Ductal Adenocarcinoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

29/29 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 98 cells / 6 specks detached; iLISI median 1.00→9.31 of 29 samples); UMAP axis ratio 1.37 (compact mass, not a strip of islands). Proliferation is not a confound (module–cycling |ρ|≤0.05).

Gate 2 — module geography → GO (non-branching)

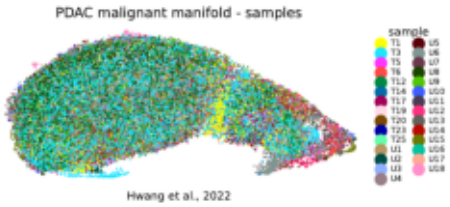
Single organized cross-repression axis, not a bifurcation: Gast↔pSom mutually exclusive (smoothed r=−0.32; joint Gast&pSom-high enrichment 0.44 — no co-expression hub). pSom high at the origin declines along pseudotime (ρ=−0.34), Gast rises (ρ=+0.29); neural crest is a terminal / inverse readout (crest-high decile mean pt=0.63), anti-correlated with both modules (vs pSom r=−0.39, vs Gast r=−0.30 — not a crest-arm inversion).

Gate 3 — hub mutual exclusivity → NA

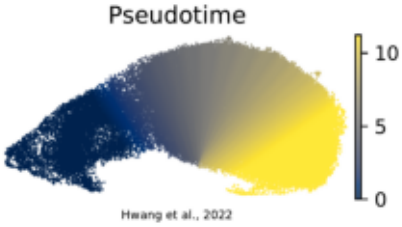
Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for competitive cross-repression. Branching-only correlation / modules steps are not produced.

STEP FIGURES

Gate 1: integration



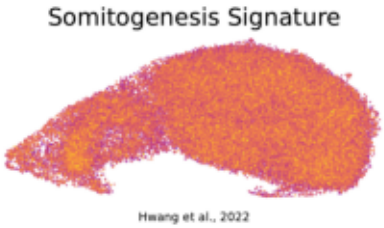
Trajectory / root



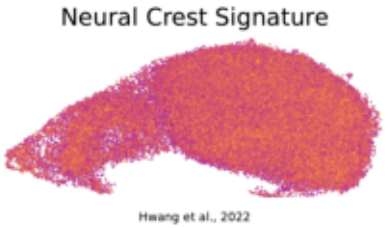
Gast signature



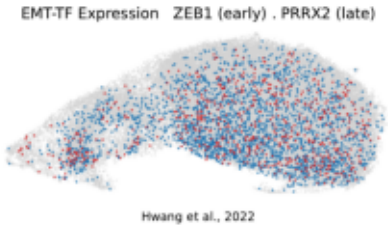
pSom signature



Crest signature



EMT-TFs



Dataset. PDAC (Hwang et al., 2022), snRNA-seq TP10K — 43 samples / ≈225k nuclei across 3 group matrices; malignant only: 64,538 nuclei → 64,104 across 29 samples ≥100 cells → 64,006 after island removal enter the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Hwang's PDAC malignant manifold is a single continuous axis rooted at a somitogenesis-high / neural-crest-low origin, with organized Gast↔pSom cross-repression and neural crest as a terminal inverse gradient — the developmental axis without Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The curve is fold-free (2 tips, 0 forks; self-approach 2.81) and continuous on an independent Harmony-PCA geometry (ρ=−0.73; centroid-path ratio 2.27). The axis spans both transcriptional compartments (Basal-like 54,795 / Classical-like 9,211) and is orthogonal to cell cycle (≈93% non-cycling), so the whole manifold is kept. Raw module means are low-amplitude (Gast −0.004 / pSom −0.002 / Crest +0.003) — as in the prior failed Hwang run — but the geography is decisive: organized cross-repression + terminal crest gradient, read off the signature embeddings, not a scalar. EMT-TFs ZEB1 (early / root) → PRRX2 (late / terminus); ZEB1 dominant (15.7% expressing).

Parameter choices

Min cells/sample	≥100 floor (29/29 kept)	**Integration**	Harmony θ=10; n_pcs=20
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Basal-like 54,795 / Classical-like 9,211
Geometry	Linear (non-branching)	**EMT-TFs**	ZEB1 (root) → PRRX2 (terminus)